

SEIS615-02 – Cloud Computing – Fall 2026

Time: 5:30–8:30pm, Thursdays (09/10-12/17)

Classroom: OWS 275

Instructor: Ron C. Chiang, Office: SCC 327

cchiang@stthomas.edu (Please put “SEIS615” in the subject line)

Office Hours: By appointments

Course Description

This course covers the fundamentals of IT infrastructure in the cloud. It provides a detailed overview of cloud concepts, services, security, architecture, and economics. This course will examine the theory behind these modern practices and the real-world implementation challenges faced by IT organizations. Students will learn how to design and implement cloud-based solutions. While the lessons will cover a number of theoretical concepts, we will primarily learn by doing. Students will gain hands-on experience with widely used IT and AI platforms.

Course Objective

- Make architectural decisions based on cloud architecting principles and best practices
- Understand how to design scalable, reliable, and highly available infrastructure
- Develop specific skills related to best practices of Cloud Architecting to increase performance and reduce cost of cloud infrastructure

Texts

- No required textbooks
- Additional material as indicated later by the instructors

Canvas

Lecture notes (powerpoint/pdf format), lecture recordings, and grades will be available on Canvas (<https://www.stthomas.edu/canvas>).

Communication Policy

Please check your St. Thomas email regularly. We will be using St. Thomas email addresses for our email communications related to this course. We will also be using Canvas extensively for this class. I will post major announcements on Canvas as the primary means of communicating en masse with the class. All lecture materials, study guides, as well as all requirements for assignments will be available on the Canvas.

Attendance Policy

Instructor will enter attendance into Canvas weekly

Course Schedule

Session	Date	Topic	Activities and Assignments
1	09/10	Cloud Concepts and Economics Cloud Architecting	Account signup Management console clickthrough Billing dashboard & Budget control
2	09/17	Cloud Security Controlling User and App Access	Lab: IAM users and groups
3	09/24	Storage Infra Dev with AI Agent	Lab: S3 Due: IAM users for developer, DB admin
4	10/01	Compute	Lab: EC2 Due: static website on S3
5	10/08	Networking	Lab: VPC Due: website on EC2
	10/22	Midterm Quiz (10/08-10/22)	Due: midterm quiz
	10/22	Midterm presentation	Due: midterm presentation files
6	10/29	Databases in Cloud	Lab: RDS Due: Design a VPC
7	11/05	Elasticity, High Availability, and Monitoring	Lab: ALB, Auto Scaling Due: RDS
8	11/12	Automating Your Architecture	Lab: CloudFormation Due: Elasticity and Availability
9	11/19	Caching and Content Delivery Microservices and Serverless Architectures	Lab: Serverless architecture Due: CloudFormation
10	12/03	Event-Driven Architecture Business Continuity and Disaster Recovery	Lab: S3, SQS, SNS
	12/10	Final presentation	Due: final presentation files
	12/17	Final Quiz (12/03-12/17)	Due: Serverless Due: final quiz

Grading Policy

The final grade for the course will be based on the following:

- Attendance (10%)
 - 50% attendance score reduction for each unexcused absence
 - Two or more absences will result in a grade reduction a report to the department
 - If you cannot make it to a class, **please notify the instructor via email prior to the class time**
- Assignments (40%)
 - Assignments due by beginning of a class
 - No credit for late submissions
- Exams (20%)
 - Based on course materials and assignments
 - Midterm Quiz (10%)
 - Final Quiz (10%)

- Presentation (30%)
 - Midterm presentation (15%)
 - Final presentation (15%)

Letter grade will be assigned as follows:

90 – 100	: A-, A
75 – 90	: B-, B, B+
60 – 75	: C-, C, C+
Below 60	: F

Recording of Classroom Activities

All recordings of class sessions using any device is expressly prohibited without the written permission of the instructor.

Students with Disabilities

Academic accommodations will be provided for qualified students with documented disabilities including but not limited to mental health diagnoses, learning disabilities, Attention Deficit Disorder, Autism, chronic medical conditions, visual, mobility, and hearing disabilities. Students are invited to contact the Disability Resources office about accommodations early in the semester. Appointments can be made by calling 651-962-6315 or in person in Murray Herrick, room 110. For further information, you can locate the Disability Resources office on the web at <http://www.stthomas.edu/enhancementprog/>.

Statement Related to Influenza and COVID19

The University of St. Thomas is committed to a healthy campus community. To help limit the spread of influenza or influenza-like illnesses, the Center for Disease Control has provided college campuses the following recommendation: students, faculty, or staff with influenza-like illnesses (temperature of 100.0 or greater, plus a cough or sore throat) are directed to self-isolate (or stay home) for at least 24 hours after their fever is gone without the use of fever-reducing medicine. In the event that students are unable to attend classes due to this self-isolation recommendation, they should inform their professors of their absence.

Academic Integrity

Academic integrity is defined as not cheating and not plagiarizing; honesty and trust among students and between students and faculty are essential for a strong, functioning academic community. Consequently, students are expected to do their own work on all academic assignments, tests, projects and research/term papers. Academic dishonesty, whether cheating, plagiarism or some other form of dishonest conduct related to academic coursework and listed in the Student Policy Book under “Discipline: Rules of Conduct” will automatically result in failure

for the work involved. But academic dishonesty could also result in failure for the course and, in the event of a second incident of academic dishonesty, suspension from the University.

Exam Proctoring Policy: “The University of St. Thomas continues to strive for academic excellence and integrity in times of continuous technological advancement. The University has selected **Honorlock**, an exam monitoring tool to assist in combating academic dishonesty in the online testing environment. This is part of a three-step approach of prevention, detection, and communication. By registering for a course within the School of Engineering’s Graduate Programs, students are agreeing to install the tools on exam-taking devices for the duration of any course exam, if required by the faculty instructor.”

Here are the common ways to violate the academic integrity code:

Cheating: Using, accessing, or trying to use unauthorized materials, information, or study aids in any academic work submitted for credit; giving or receiving unauthorized materials or aid in an academic exercise.

Fabrication or misrepresenting: Falsifying or inventing any information or citation in an academic exercise; fabricating or misrepresenting a reason for an absence or missed deadline; submitting academic work completed largely in one class to satisfy an assignment in a second class without instructor permission; fabricating or tampering with academic records.

Facilitating Academic Dishonesty: Intentionally or knowingly helping or attempting to help another to violate a provision of the institutional code of academic integrity.

Plagiarism: Claiming or submitting the ideas, words, or statements of another person as one's own without proper citation regardless of intent to deceive. Even inadvertent plagiarism is a serious matter; students must familiarize themselves with citation standards.

In cases of violating the academic integrity code, the instructor will impose a minimum sanction of failure of work involved. The instructor will inform the student and the program director in writing of:

1. the nature of the offense,
2. the penalty imposed within the course;
3. the recommendation of the instructor as to whether further disciplinary action by the director is warranted.

If the instructor or the director of the program determines that further disciplinary action is warranted, a disciplinary hearing shall be commenced at the request of either the instructor or the director. (If there is a previous offense of this nature on the student’s record, a hearing is mandatory.)

Disclaimer

This syllabus is not a contract, but a plan for action. The instructors reserve the right to alter its stipulations, upon prior notification to students, if and when educational circumstances warrant changes.